

01_Week_8_Proposal

1. Study Group and Project

Study Group: GRP6

Members:

- Ralph Aron Bonifacio
- Andrei Karl Dela Cruz
- Eivnard Seraphim Franco
- Matthew Zachary Medallon
- Josef Jayden Nepomuceno

Project Title:

Analyzing Palay and Corn Production and Harvested Area Across Philippine Regions, 2016–2026

Intended Audience:

Philippine agricultural planners, policymakers, researchers, and other stakeholders interested in regional palay and corn production.

Problem:

This project examines changes in palay and corn production and harvested area across Philippine regions from 2016 to 2026. It also examines the relationship between harvested area and production to provide a clearer view of regional agricultural patterns.

Analytical Questions:

1. How has the production of palay and corn changed across Philippine regions from 2016 to 2026?
2. How has the area harvested for palay and corn changed across Philippine regions from 2016 to 2026?
3. How does harvested area relate to the production of palay and corn across regions and crop types from 2016 to 2026?

2. Proposed Datasets

Dataset 1: Volume of Production

- **Provider:** Philippine Statistics Authority (PSA) - OpenSTAT
- **Dataset:** *Palay and Corn: Volume of Production in Metric Tons by Ecosystem/Croptype, Quarter, Semester, Region and Province, 1987–2026*
- **Source URL:**
https://openstat.psa.gov.ph/PXWeb/pxweb/en/DB/DB_2E_CS/0012E4EVCP0.px/
- **Usage Conditions:** The dataset is publicly available through PSA OpenSTAT. The source will be properly cited in the project.
- **Coverage:** 1987–2026
- **Unit:** Metric tons

- **Relevant Fields:** Ecosystem/Croptype, Geolocation, and the quarterly, semester, and annual period values
- **Likely Integration Keys:** Ecosystem/Croptype + Geolocation

Dataset 2: Area Harvested

- **Provider:** Philippine Statistics Authority (PSA) - OpenSTAT
- **Dataset:** *Palay and Corn: Area Harvested in Hectares by Ecosystem/Croptype, Quarter, Semester, Region and Province, 1987–2026*
- **Source URL:**
https://openstat.psa.gov.ph/PXWeb/pxweb/en/DB/DB_2E_CS/0022E4EAHC0.px/
- **Usage Conditions:** The dataset is publicly available through PSA OpenSTAT. The source will be properly cited in the project.
- **Coverage:** 1987–2026
- **Unit:** Hectares
- **Relevant Fields:** Ecosystem/Croptype, Geolocation, and the quarterly, semester, and annual period values
- **Likely Integration Keys:** Ecosystem/Croptype + Geolocation

Dataset Relationship

Both datasets contain the same Ecosystem/Croptype and Geolocation fields, which will be used as the proposed integration keys. The initial validation found **654 unique key combinations** in each dataset, with all **654 combinations matching** between the two datasets.

Although the original datasets cover 1987–2026, the proposed analysis will focus on **2016–2026**.

3. Initial Python Inspection

Before starting the analysis, the group first inspected the two datasets in Python to understand their structure and identify issues that may need to be addressed. The Production and Area Harvested files were loaded using `pd.read_csv()`. The `header=1` option was used because the first row contains the dataset title, while the following row contains the actual column names. The first few rows, dataset size, column names, and data types were then checked to make sure the files were loaded properly.

Both datasets contain 654 rows and 282 columns, with data covering 1987 to 2026. The first two columns are *Ecosystem/Croptype* and *Geolocation*, followed by quarterly, semester, and annual values for each year. All 282 columns were initially read as strings. No actual *NaN* values were found, but both datasets contain `..` entries in some period columns. The Production dataset contains 17,253 such entries, while the Area Harvested dataset contains 17,262. The inspection showed that some of these entries occur in later 2026 periods, such as Quarter 3, Quarter 4, Semester 2, and Annual, where no values are currently present in the downloaded data. These entries will need to be handled during the cleaning stage before the data is used for numerical analysis.

The inspection then focused on the fields that could be used to connect the two datasets. Both contain the same six *Ecosystem/Croptype* values: Irrigated Palay, Rainfed Palay, Palay, White Corn, Yellow Corn, and Corn. The *Geolocation* field also contains 109 unique values in each dataset. Because these fields are present in both datasets, their combination was checked as a possible way to match the records.

This produced 654 unique combinations in each dataset. A comparison of the two sets of combinations showed that all 654 were present in both datasets, with no unmatched combinations found. This provides

an initial basis for using *Ecosystem/Croptype* and *Geolocation* when the datasets are combined in a later stage.

The inspection also helped clarify how the datasets will support the three analytical questions. The Production dataset will be used to examine changes in palay and corn production across regions, while the Area Harvested dataset will be used to examine changes in harvested area. Both will be used for the third question, which considers the relationship between harvested area and production across regions and crop types. Although the original datasets cover 1987 to 2026, the proposed analysis will be limited to 2016 to 2026 to keep the project focused. The identified data issues and matching fields will then be addressed further during the cleaning and integration stages.

4. Proposed Components

To ensure a robust and reproducible workflow, the Python pipeline is divided into distinct, modular components. Each phase features a primary programmer responsible for implementation and a designated peer reviewer to enforce code quality and logic.

<i>Acquisition and Validation:</i>	Bonifacio will lead data acquisition, initial inspection, and validation, ensuring repeatable programmatic loading and basic quality checks. Nepomuceno will act as his peer reviewer.
<i>Cleaning and NumPy Execution:</i>	Data cleaning is split to efficiently handle multiple sources. Dela Cruz will manage the production dataset's cleaning and preprocessing, reviewed by Bonifacio. Franco is tasked with cleaning the area harvested dataset and executing the required NumPy numerical operations, with Dela Cruz providing the peer review.
<i>Integration and Analysis:</i>	Medallion will drive dataset integration and data analysis. He will merge the cleaned tables and compute the derived measures needed to answer the project's analytical questions, while Franco reviews the merge cardinality and analytical validity.
<i>Visualization, Testing, and Reliability:</i>	Nepomuceno will finalize the output through data visualization, automated testing, and reliability checks, generating purposeful figures and necessary assertions. Medallion will serve as his peer reviewer to confirm the visual interpretations and testing framework.

Every member will contribute substantive Python code and review another member's work. Responsibilities may be adjusted as the project progresses based on operational needs and instructor feedback.

5. Likely Risks

Access Risks - There could be downtime on the PSA OpenSTAT portal, API rate limits, or unexpected changes in file download structures that could interrupt automated data retrieval or block member access. To mitigate this, **the group** has downloaded static files of both raw datasets and committed them to the data/raw/ folder in the repository so **the group** can work offline. **The group's** feasible fallback is to maintain a secondary backup folder for the raw files, accessible to all group members, or to adjust **the group's** coverage by a year if entries remain incomplete.

Mismatched Keys - Inconsistency in composite key formatting across datasets, like padding, regional namings, or provincial reclassifications that emerge after cleaning and cause missing values during integration, to mitigate, **the group** will apply text normalization prior to merging and enforce key alignment assertions in the pipeline. **The group's** feasible fallback is to aggregate both datasets to a higher-level regional * annual scale before merging to bypass lower-level key mismatches.

Missing Information - The presence of missing values and zero production entries due to localized crop damage, unrecorded quarters or non-numeric placeholders in historical OpenSTAT sheets, to mitigate, **the group** will configure explicit non-numeric parsing tools upon reading files and establish clear logic to differentiate legitimate zero-yield occurrences from unrecorded data. **The group's** feasible fallback is to impute minor missing values using seasonal regional averages or aggregate quarterly records into annual totals to minimize data gaps.

Privacy - Unintended inclusion or handling of sensitive data, because PSA OpenSTAT provides macro-level agricultural statistics, the Personally Identifiable Information (PII) is not present; however, misattribution or misinterpretation of public figures poses compliance risks. To mitigate this, **the group** will ensure full compliance with PSA OpenSTAT public domain usage guidelines by making proper citations and documenting all steps. **The group's** feasible fallback is to publish a dedicated data lineage and methodology document in the docs/ folder to maintain full transparency.

Scope - **The group's** scope covers the tracking of two distinct crops across multiple regions and ecosystems. Provinces and quarters over a 10-year period, leading to bloated notebooks, **the group** will mitigate this by restricting the objectives to the three core questions defined in **the group's** proposal and prioritizing regional and national trends over provincial trends. **The group's** feasible fallback is to standardize all primary visualizations to Regional Annual totals and shift provincial or ecosystem breakdowns to an optional appendix.

Schedule - Sequential Task dependences among group members may lead to workflow delays. To mitigate this, **the group** will generate synthetic integrated datasets in **the group's** data/samples/ folder so that visualization, testing, and analysis can proceed in parallel while cleaning scripts are finalized. **The group's** feasible fallback is to reassign lagging pipelines to the assigned peer reviewers according to the group responsibility matrix to meet the deadline.

6. Workplan

Week	Planned Activities	Expected Output
Week 8	Finalize project scope, analytical questions, datasets, initial inspection, integration keys, responsibilities, and proposal.	Week 8 proposal, initial inspection code, sample data, and repository commit
Week 9	Review Week 8 feedback and revise the proposal. Inspect both datasets more thoroughly and begin the necessary cleaning and standardization.	Revised proposal, cleaning plan, and initial cleaning code
Week 10	Complete the necessary cleaning and validation for both datasets. Check data types, missing values, duplicates, labels, and the 2016–2026 scope.	Cleaned and validated datasets, quality-check results, and validation code
Week 11	Integrate the production and harvested-area datasets using the approved integration keys. Check for unmatched records, duplicates, and unexpected results.	Integrated dataset, integration code, and integration validation results

Week 12	Begin exploratory analysis of the integrated dataset. Examine production and harvested-area patterns across regions, crop types, and years and identify relevant trends for further analysis.	Exploratory analysis results and initial tables
Week 13	Conduct the main analysis for production and harvested-area trends across regions, crop types, and years.	Q1 and Q2 analysis results
Week 14	Analyze the relationship between harvested area and production. Refine and verify the results from all three analytical questions.	Completed Q3 analysis and finalized analysis results
Week 15	Develop and refine visualizations that directly address the three analytical questions.	Final visualization outputs and visualization code
Week 16	Conduct testing and reliability checks. Run the complete workflow from raw data through integration, analysis, and visualization. Fix identified issues.	Tested workflow, corrected code, and reliability-check results
Week 17	Finalize documentation, presentation, repository, outputs, and required project files.	Final project package, repository, and presentation

Questions for Instructor Feedback

- During our initial inspection, we found .. entries in several period columns. Some of these appear in later 2026 periods, such as Quarter 3, Quarter 4, Semester 2, and Annual, where no values are currently present in the downloaded data. Since these entries are not detected as NaN, should they be converted to NaN during cleaning, or is there another approach that should be used?
- Our datasets contain quarterly, semester, and annual values for each year from 2016 to 2026. Since these different time levels provide different ways of looking at the data, should we consider all three in the analysis, or would it be better to focus on one time level, such as the annual values, for a more consistent comparison?
- During our initial validation, we found 654 unique combinations of Ecosystem/Croptype and Geolocation in each dataset, with all 654 combinations matching between the two datasets. Aside from checking for unmatched keys, what other validation checks would be important to perform after integrating the datasets?
- Since the main data processing will be done using pandas, we are considering using NumPy for a derived calculation such as production per hectare. Would this count as a meaningful use of NumPy for the project requirement?
- For the required three purposeful figures, we are planning to relate each figure to one of our three analytical questions: production trends, harvested-area trends, and the relationship between production and harvested area. Would this approach be appropriate for the visualization requirement?
- Based on our initial inspection and proposed approach, are there any concerns with our current dataset scope, integration approach, or use of the PSA OpenSTAT data that we should address before proceeding with the next stages of the project?